

Improve Design Efficiency and Test Capabilities with HIL Simulation

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Abstract - Customer expectations and governmental requirements have changed the way engineers develop products. From automobiles and airplanes to industrial equipment and national defense systems, designers are adding intelligence to their products in the form of electronic controllers. A direct result of this evolution of product design; the testing complexity for these products is growing at an exponential rate. To address this challenge, many engineers have turned to a technique called Hardware-in-the-Loop (HIL) Simulation. HIL Simulation is a test technique that allows the engineer to begin testing their electronic controller earlier in the development process and with greater flexibility compared to physical testing alone. While not a replacement for physical testing; in many situations, such as the testing of a flight control system, HIL Simulation is the only viable option for development testing due to the potential consequences that may result from a failed test. In this paper, we will show how HIL Simulation is being used today and discuss a general architecture for building a HIL Simulator. We will also comment on many considerations that should be taken when specifying a HIL Simulator.

Keywords – simulation, hardware in the loop, validation, hil

I. WHAT IS HIL SIMULATION?

HIL simulation is a dynamic test technique that simulates the I/O behavior of a physical system that interfaces to an electronic controller in real-time. It is dynamic because the values of stimulus signals generated by a simulator are a function of an electronic controller's response from the previous cycle. Other variables such as test profiles and in-line analysis results may also influence the calculation of stimulus values, but it is their dependence on the response from a unit under test that differentiates HIL simulation from other test techniques.

II. HOW DOES HIL SIMULATION WORK?

In a closed-loop control system, the current state of the system being controlled is fed back to the electronic controller through sensor measurements. The device uses these measurements to help determine the appropriate actuator values in order to attain a desired operating condition for the system (see Fig. 1).

To understand how this is accomplished, let's first consider

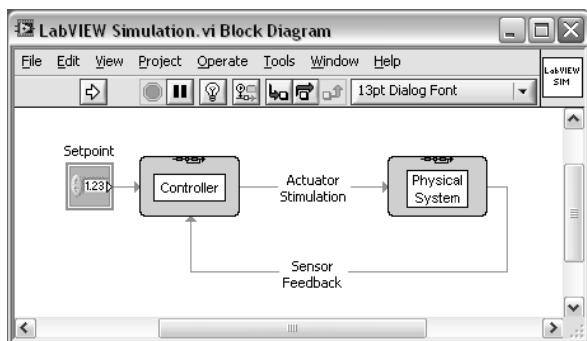


Fig. 1 – Closed-Loop Control System

what an electronic controller “knows” about the world around it. A typical controller unit consists of an embedded computer system with integrated electronics for sensor and actuator signal conditioning and digital communication protocols. Therefore, an accurate representation of the voltage, current, impedance, and timing characteristics of the physical system being controlled is indistinguishable from the actual system. However, a HIL simulator must meet certain requirements in order to accurately represent a physical system.

A HIL simulator must be able to generate and acquire signals at the same amplitude and rate of change that a physical system would produce. To accurately represent how different operating conditions affect a controller electrically, a HIL simulator must create the impedances, or loads, that an electronic controller experiences in real-world operation. The impedance seen by the outputs determines the amount of current drawn from the device. Although the electronic controller hardware may meet all design requirements, faults external to the device or unexpected control algorithm results can produce actuator command signals outside of the hardware specification that may result in incorrect system behavior. In order to identify such issues, the impedance characteristics of a system must also be simulated in order to produce an accurate simulation. Finally, there are a variety of standardized communication systems used in flight systems to distribute data between sensors, actuators, and controllers such as ARINC-429 and MIL-STD-1553/1773. As flight systems evolve, there are also a growing number of custom digital protocols being used to implement communications in the control systems.

In addition to being able to interface electrically with an electronic controller, a HIL simulator must determine the correct values to be produced in order to exercise or test the device. State charts, programming languages, and dynamic system models are commonly used to represent the I/O behavior (dynamic response) of a physical system. However, it is critical that a HIL simulator also be able to produce these values with accurate timing characteristics. This typically requires a real-time operating system to ensure all signals can be updated at a rate that will preserve the realistic representation of a physical system in time. Depending on the complexity of a system being simulated and the fidelity of its representation, parallel processing techniques such as FPGAs, multicore processors, and deterministic distributed processing interfaces may be necessary to complete output response calculations while maintaining timing accuracy.

III. WHY IS HIL SIMULATION IMPORTANT

The increasing complexity inherent in flight systems works against business pressures to reduce development cost and increase product quality. Fortunately, HIL simulation has proven to be a practical solution to these diverging challenges. While HIL simulation does not replace the need for physical testing, it does enable engineers to tests earlier in the development cycle providing the following benefits:

- Make educated design decisions – evaluate specific design alternatives early in the development process based on test data
- Resolve errors more efficiently – identify design errors earlier when they are less expensive to correct
- Achieve greater testing capacity – enable “lights-out” testing of systems where unmonitored physical testing is not possible
- Increase test coverage – test embedded controllers under extreme conditions that might not be practical for physical testing due to safety or equipment damage concerns
- Increase test repeatability – isolate deficiencies in electronic controllers even if they occur only under certain circumstances

IV. IMPORTANT HIL SYSTEM CONSIDERATIONS

There are many options for implementing HIL simulation as part of your test system. However, to ensure the short- and long-term success of your investment, it must be both flexible and open.

With the drive to continually reduce the cost of test, a flexible solution is essential to making HIL simulation practical in your development process. The ability to test a variety of embedded controller configurations without downtime or the need for specialized services keeps the cost of ownership low. Furthermore, an effective HIL simulation solution should be able to rapidly adapt to changes encountered during and between development cycles. A small change in the test process or configuration should not require a major renovation of your HIL simulator.

As technologies advance in embedded control technologies, new communication buses, vision inspection systems, RF instrumentation, and other specialized instruments make implementation of HIL simulation test systems necessary. At the current rate of innovation, it is not practical to expect a single vendor to simultaneously meet the time-to-market, quality, and cost expectations for all of the latest technologies. An open HIL simulation solution ensures that you will always be able to integrate the technologies required to test your device.

V. THE FUTURE

Business and technical challenges will continue to grow in the aerospace industry. As technology evolves to address these challenges, the role of HIL simulation in control system development will become increasingly important. By enabling tests earlier in the development cycle and removing the limitations of physical test, HIL simulation is helping to reduce development cost and increase the quality of embedded control systems in the face of these challenges.